- Part 1: Quantum Gates – Quick Refresher
    - Single-qubit gates
    - Two-qubit gate – CNOT
    - Toffoli Gate (CCNOT – Controlled-Controlled-NOT)
    - Building an Oracle (U_f) with Toffoli
    - SWAP Gate
    - Fredkin Gate (CSWAP – Controlled-SWAP)
    - Summary: Toffoli, SWAP, Fredkin
    - Part 2: Decomposition and Universality of Quantum Gates
        - Universality of a set of quantum gates
        - Kitaev's Circuit Simulation Theorem (Solovay–Kitaev)
- Part 2: The Quantum Algorithms – Oracle Model
    - The Oracle (U_f)
    - Phase kickback – the key trick
    - Deutsch Algorithm (1-bit function)
        - Deutsch–Jozsa Algorithm (n-bit function)
        - Summary – Why is this remarkable?
    - Bernstein–Vazirani Algorithm
        - A concrete 2-bit example ((s = 10))
    - Simon's Problem and Algorithm
        - A concrete 2-bit example ((s = 11))
        - The full Simon's algorithm analysis
    - Summary of all four algorithms
    - Next Lecture Preview

Lecture Overview

1. **Quantum Gates Refresher** – Quick review
2. **The Quantum Oracle Model** – What is an oracle?
3. **Deutsch-Jozsa Algorithm** – Start with **examples first**, then the general rule
4. **Bernstein-Vazirani Algorithm** – Learn a hidden string with one query
5. **Simon's Problem** – Introduction to exponential speedup
6. **AI Tool Demo** – Generating oracles with AI

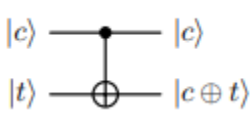
Part 1: Quantum Gates – Quick Refresher

Single-qubit gates

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}, \quad X = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \quad Z = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$$

- **Hadamard (H)**: creates superposition
  $H|0\rangle = |+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}},\ H|1\rangle = |-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$
- **Pauli-X**: bit flip (NOT)
- **Pauli-Z**: phase flip

Two-qubit gate – CNOT



If control = $|1\rangle$, flip target; else do nothing.

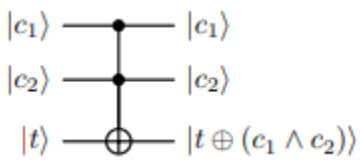
What happends when we have controlled-Z gate?
what is the output function

---

what happens when we have $|t\rangle = |+\rangle or |-\rangle$?

Toffoli Gate (CCNOT – Controlled-Controlled-NOT)

The Toffoli gate has **two control qubits** and **one target qubit**. The target flips only if **both** controls are $|1\rangle$.



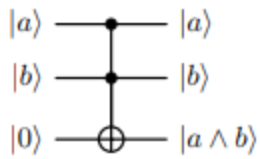
Truth table:

c_1	c_2	t	t'
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Matrix form (8×8, basis $|000\rangle, |001\rangle, \ldots, |111\rangle$):

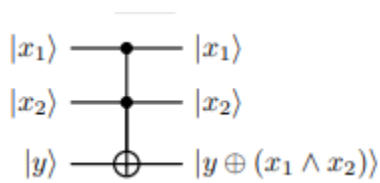
$$\text{Toffoli} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Why it matters: Toffoli is **universal for classical reversible computation** – any classical Boolean circuit can be built from Toffoli gates (plus ancilla bits). In particular, it can compute the AND of two bits into a target initially $|0\rangle$:

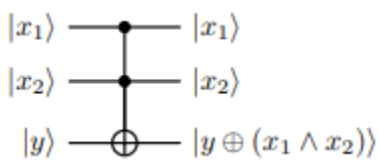


Building an Oracle U_f with Toffoli

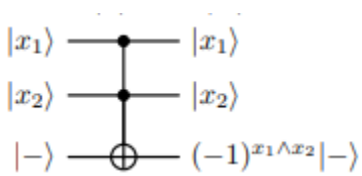
A classical Boolean function $f : \{0,1\}^n \to \{0,1\}$ can be expressed as a circuit of AND, OR, NOT gates. Since Toffoli can compute AND (with a target initially $|0\rangle$) and CNOT can copy bits, we can construct a reversible circuit for f using Toffoli and CNOT. This circuit implements the unitary:



Example: $n = 2$, $f(x) = x_1 \wedge x_2$ (AND). Using Toffoli:



Phase oracle from Toffoli: To get the phase oracle $|x\rangle \to (-1)^{f(x)}|x\rangle$, we replace $|y\rangle$ with $|-\rangle$:



Because of phase kickback, the target $|-\rangle$ remains $|-\rangle$ and the controls acquire a phase $(-1)^{x_1 \wedge x_2}$.

General procedure: Any classical circuit for f using AND, OR, NOT can be turned into a reversible circuit using Toffoli and CNOT. Then setting the output qubit to $|-\rangle$ gives the phase oracle.

SWAP Gate

The SWAP gate exchanges the states of two qubits.

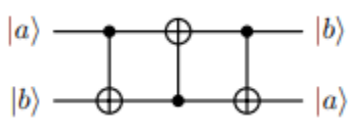
Truth table:

a	b	a'	b'
0	0	0	0
0	1	1	0
1	0	0	1
1	1	1	1

Matrix form (4×4, basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$):

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Decomposition using three CNOTs:

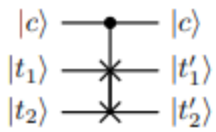


Verification:

1. CNOT(a→b): $|ab\rangle \to |a, a \oplus b\rangle$
2. CNOT(b→a): $|a, a \oplus b\rangle \to |a \oplus (a \oplus b), a \oplus b\rangle = |b, a \oplus b\rangle$
3. CNOT(a→b): $|b, a \oplus b\rangle \to |b, b \oplus (a \oplus b)\rangle = |b, a\rangle$

Fredkin Gate (CSWAP – Controlled-SWAP)

The Fredkin gate has **one control qubit** and **two target qubits**. If control = $|1\rangle$, the two targets are swapped; if control = $|0\rangle$, they remain unchanged.



Truth table:

c	t_1	t_2	t_1'	t_2'
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	1	1
1	0	0	0	0
1	0	1	1	0
1	1	0	0	1
1	1	1	1	1

Matrix form (8×8):

$$\text{Fredkin} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Relation to SWAP: If control = $|1\rangle$, Fredkin acts as SWAP on the two targets. If control = $|0\rangle$, it acts as identity. Thus Fredkin = controlled-SWAP.

Universality: Like Toffoli, Fredkin is also **universal** for **classical** reversible computation and has the additional property of **conserving the number of 1's** (Hamming weight preserving).

PennyLane code:

```
import pennylane as qml
```

```python
dev = qml.device('default.qubit', wires=3)
@qml.qnode(dev)
def toffoli_demo(a, b, t):
    if a: qml.PauliX(0)
    if b: qml.PauliX(1)
    if t: qml.PauliX(2)
    qml.Toffoli(wires=[0,1,2])
    return qml.state()
print("Toffoli(1,1,0):", toffoli_demo(1,1,0))  # target becomes 1
```

```python
dev = qml.device('default.qubit', wires=2)
@qml.qnode(dev)
def swap_demo(a, b):
    def circuit():
        if a: qml.PauliX(0)
        if b: qml.PauliX(1)
        qml.SWAP(wires=[0,1])
        return qml.state()
    return circuit()

print("SWAP(1,0):", swap_demo(1,0))
```

```
dev = qml.device('default.qubit', wires=3)
@qml.qnode(dev)
def fredkin_demo(c, t1, t2):
    if c: qml.PauliX(0)
    if t1: qml.PauliX(1)
    if t2: qml.PauliX(2)
    qml.CSWAP(wires=[0,1,2])   # CSWAP = Fredkin
    return qml.state()

print("Fredkin(1,1,0):", fredkin_demo(1,1,0)) # swap → t1=0, t2=1
```

Summary: Toffoli, SWAP, Fredkin

- **Toffoli** computes AND and is universal for reversible classical computation. It can be used to build oracles for any Boolean function.
- **SWAP** exchanges two qubits and can be decomposed into three CNOTs.
- **Fredkin** is a controlled-SWAP, also universal and preserves Hamming weight.

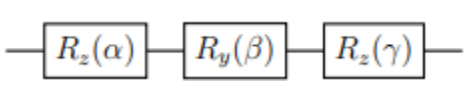
These gates are essential building blocks for quantum circuits, especially when implementing classical functions as oracles in algorithms like Deutsch-Jozsa, Grover, and Shor.

Part 2: Decomposition and Universality of Quantum Gates

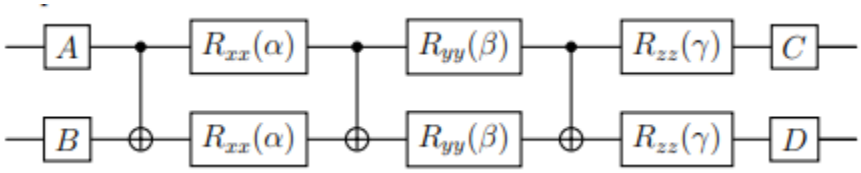
Idea: Any complex quantum gate can be written as a sequence of simpler gates.

Single Qubit: Euler Angles

Any single-qubit unitary (upto global phase) $U = R_z(\alpha)R_y(\beta)R_z(\gamma)$



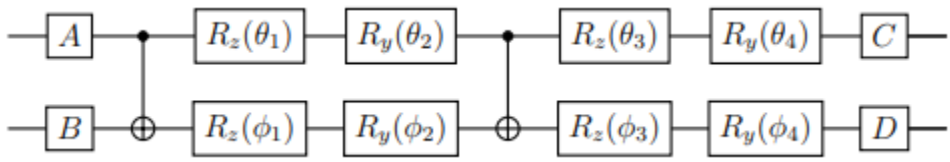
Two Qubits: Cartan Decomposotion



- A, B, C, D are single-qubit gates

---

alternative decomposition (Kraus-Cirac decomposition)



Why Does This Matter?

- **Real quantum computers** only implement a small set of native gates (e.g., CNOT + single-qubit rotations)
- **Compilers** can automatically decompose complex gates into allowed ones

---

Universality of a set of quantum gates

- A set of gates is **universal** if we can build **any possible quantum circuit** using only those gates
- Like NAND gates in classical computing – but quantum gates need to handle **continuous rotations**

{H, S, CNOT} (Clifford Gates) – Not Universal!

- Gates: H, $S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$, CNOT
- These can only map basis states to basis states (up to signs)
- Cannot create arbitrary rotations → **not universal**

Why not?

- This set generates the **Clifford group**.
    - Clifford gates map Pauli operators to Pauli operators under conjugation (e.g., $HXH = Z$, $SXS^\dagger = Y$, etc.).
    - The Clifford group on n qubits is a **finite group** (size $\sim 2^{n^2+2n}$).

- Any circuit built only from H, S, and CNOT can only produce a finite set of unitary operations.
    - You cannot create states like $|0\rangle + e^{i\pi/4}|1\rangle$ (an eigenstate of the T gate) exactly or even approximate it arbitrarily well.
    - Without a non-Clifford gate (e.g., T, or a rotation by an irrational angle), you cannot achieve universality.

What *is* needed for universality?

A common universal gate set is

- **{H, T, CNOT}**
- or **{H, S, T, CNOT}**.

> Adding any gate outside the Clifford group (e.g., $T = \sqrt{S}$) breaks the finite-group restriction and allows approximation of any unitary via the Solovay–Kitaev theorem.

Summary Table

Gate Set	Universal?	Notes
CNOT + all single-qubit rotations	✅ Yes (exact)	Most common theoretical set
CNOT + H + T	✅ Yes (approximate)	Standard discrete universal set
Toffoli + H + T	✅ Yes	Also universal
Toffoli + H	❌ No	Missing non-Clifford rotation (e.g., T)
H + S + CNOT (Clifford)	❌ No	Only generates a finite group
NAND (classical)	✅ Yes (classical)	Only for bits, not qubits

Key Takeaway

> **Universality** requires either **all continuous single-qubit rotations** or a **discrete set containing a non-Clifford gate** like T.
> **Toffoli + H** is not enough – you need an extra ingredient.

Kitaev's Circuit Simulation Theorem (Solovay–Kitaev)

The Big Picture

Suppose you have a **finite set of basic quantum gates** (like Hadamard, T, and CNOT) that can eventually create any quantum operation.
You want to build a specific unitary gate U (e.g., a rotation by 1 degree).
The theorem tells you: **you can approximate U to very high accuracy using surprisingly few gates from your set.**

The Trade-off: Error vs. Number of Gates

Let:

- ε = the maximum error you allow (e.g., $\varepsilon = 0.001$).
- L = the number of gates in the sequence that approximates U.

The theorem states that you can always find a sequence with

$$L = O\left(\log^c \frac{1}{\varepsilon}\right)$$

where c is a small constant (typically $c \approx 1$–4 for practical cases).

Key point

- **Exponential improvement in error requires only a polynomial (in fact, polylogarithmic) increase in gate count.**
  To make the error ten times smaller, you only need to add a *constant* number of extra gates (because $\log(1/\varepsilon)$ grows very slowly).
- **Example:**
  If a certain approximation uses 1000 gates to achieve error 10^{-3}, then to achieve error 10^{-6} (1000 times smaller) you might need only about $1000 \times \log^c(1000) \approx$ a few thousand gates, not a million.

Why is this surprising?

Without the theorem, you might fear that reducing error by a factor of k would require k times more gates (linear overhead).
Kitaev (and later Solovay) proved that quantum gates are much more efficient: you can “zoom in” on the desired operation using a recursive, self-similar construction – similar to how a fractal fills space with a few simple rules.

Formal Statement (for the interested student)

> **Solovay–Kitaev Theorem**
> Let G be a finite set of gates that is universal for $SU(d)$ (i.e., any unitary can be approximated arbitrarily well).
> Then there exists a constant c and an algorithm that, for any target $U \in SU(d)$ and any $\varepsilon > 0$, produces a sequence of gates from G of length L such that
>
> $$\|U - V\| \leq \varepsilon \quad \text{and} \quad L = O\left(\log^c \frac{1}{\varepsilon}\right).$$

Here $\|\cdot\|$ is the operator norm (distance between unitaries).

Take-home message for quantum computing

- **You don't need an infinite hardware instruction set.** A few basic gates are enough to build any quantum circuit efficiently.
- **The overhead for high precision is very mild** – a key reason why fault-tolerant quantum computing is possible in theory.

Below is a **complete, step-by-step teaching module** that builds from the basic oracle model to the Deutsch and Deutsch–Jozsa algorithms. It assumes the student has seen basic quantum circuits and superposition.

Part 2: The Quantum Algorithms – Oracle Model

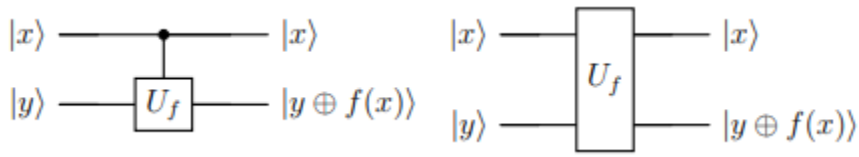
The Oracle U_f

A quantum oracle is a black box that implements a function $f : \{0,1\}^n \to \{0,1\}$ as a unitary transformation:

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

Here x is n qubits (the input register), y is a single qubit (the output register), and $\oplus$ is addition modulo 2.

Circuit representation (as controlled operaiton or more generic way)



Phase kickback – the key trick

If we prepare the output qubit in the state $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$, a remarkable thing happens:

$$\begin{aligned}
U_f|x\rangle|-\rangle &= U_f|x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}} \\
&= \frac{1}{\sqrt{2}}\big(U_f|x\rangle|0\rangle - U_f|x\rangle|1\rangle\big) \\
&= \frac{1}{\sqrt{2}}\big(|x\rangle|0 \oplus f(x)\rangle - |x\rangle|1 \oplus f(x)\rangle\big) \\
&= \frac{1}{\sqrt{2}}|x\rangle\big(|f(x)\rangle - |1 - f(x)\rangle\big)
\end{aligned}$$

$$\frac{1}{\sqrt{2}}|x\rangle\big(|f(x)\rangle - |1 - f(x)\rangle\big)$$

Now examine the two possibilities:

- If $f(x) = 0$: $|f(x)\rangle = |0\rangle$, $|1 - f(x)\rangle = |1\rangle \to \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$
- If $f(x) = 1$: $|f(x)\rangle = |1\rangle$, $|1 - f(x)\rangle = |0\rangle \to \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle$

We can then write:

$$U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$$

Interpretation: The output qubit stays $|-\rangle$ and a **phase** $(-1)^{f(x)}$ is kicked back onto the input register.

From now on we can ignore the output qubit and simply write:

$$\text{Oracle effect: } |x\rangle \xrightarrow{U_f} (-1)^{f(x)}|x\rangle$$

Deutsch Algorithm (1-bit function)

We are given a function $f : \{0, 1\} \to \{0, 1\}$. It is **promised** to be either:

- **Constant**: $f(0) = f(1)$ (both 0 or both 1), or
- **Balanced**: $f(0) \neq f(1)$ (one 0, one 1).

Goal: Determine which with only **one** query to the oracle.

Classical difficulty

Classically you need two queries: check $f(0)$ and $f(1)$.

Quantum implementation

One query suffices. How?

---

1. **Initial state:** $|0\rangle|1\rangle$
2. **Apply H to both qubits:**

$$|0\rangle|1\rangle \xrightarrow{H\otimes H} \left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right) = |+\rangle|-\rangle$$

---

3. **Apply oracle U_f (phase kickback):**
   The input register $|x\rangle = |+\rangle$ becomes:

$$\frac{1}{\sqrt{2}}\big((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\big)|-\rangle$$

---

4. **Apply Hadamard on the first qubit:**
   Recall $H|0\rangle = |+\rangle,\ H|1\rangle = |-\rangle$. So:

$$H\left(\frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}\right) = \frac{(-1)^{f(0)}|+\rangle + (-1)^{f(1)}|-\rangle}{\sqrt{2}}$$

   Rearranging:

$$= \frac{(-1)^{f(0)} + (-1)^{f(1)}}{2}\,|0\rangle \ + \ \frac{(-1)^{f(0)} - (-1)^{f(1)}}{2}\,|1\rangle$$

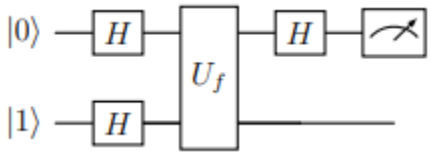
---

5. **Measurement of the first qubit:**
   - **Constant** case: $f(0) = f(1) \Rightarrow (-1)^{f(0)} = (-1)^{f(1)}$.
     Then amplitude of $|1\rangle$ = 0, amplitude of $|0\rangle$ = ± 1.
     → **Always measure** $|0\rangle$.
   - **Balanced** case: $f(0) \neq f(1) \Rightarrow (-1)^{f(0)} = -(-1)^{f(1)}$.

Then amplitude of $|0\rangle$ = 0, amplitude of $|1\rangle$ = ± 1.
$\rightarrow$ **Always measure** $|1\rangle$.

Conclusion: One measurement tells us the answer with certainty.

Quantum circuit



(Here the second qubit starts in $|1\rangle$ and is put into $|-\rangle$ by the first H. $|-\rangle$ is used for phase kickback.)

Deutsch–Jozsa Algorithm (n-bit function)

Now $f : \{0,1\}^n \rightarrow \{0,1\}$ with the promise that f is either **constant** (all outputs equal) or **balanced** (exactly half the inputs give 0, half give 1).
Goal: Decide constant vs. balanced with **one** quantum query (classically you may need $2^{n-1}+1$ queries).

Step-by-step

1. **Initial state:** $|0\rangle^{\otimes n}|1\rangle$

---

2. **Apply $H^{\otimes n}$ on the top register and H on the bottom:**
   Top becomes $\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle$, bottom becomes $|-\rangle$.

$$|\psi_0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|-\rangle$$

3. **Apply oracle U_f (phase kickback):**

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle |-\rangle$$

4. **Apply $H^{\otimes n}$ to the top register:**
   The Hadamard transform on an n-qubit basis state:

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle$$

   where $x \cdot z = \sum_{i=1}^n x_i z_i \mod 2$ (bitwise dot product).

Thus:

$$H^{\otimes n}|\psi_1\rangle = \frac{1}{2^n} \sum_{x,z} (-1)^{f(x)+x \cdot z} |z\rangle$$

The amplitude of the **all-zero** state $|z = 0\rangle$ is:

$$\frac{1}{2^n} \sum_x (-1)^{f(x)}$$

5. **Measure the top register in the computational basis.**
   - **If f is constant**: $(-1)^{f(x)} = \pm 1$ for all x.
     Then amplitude of $|0\rangle^{\otimes n} = \frac{1}{2^n} \sum_x (\pm 1) = \pm 1$.
     All other amplitudes are 0.
     $\rightarrow$ **We measure $|0\rangle^{\otimes n}$ with certainty**.

- **If f is balanced**: Exactly half the x give $+1$, half give -1.
  Then $\sum_x (-1)^{f(x)} = 0$, so amplitude of $|0\rangle^{\otimes n} = 0$.
  The probability to measure $|0\rangle^{\otimes n}$ is zero.
  → **We measure some non-zero bitstring**.

Conclusion:

- Measure $|0\ldots0\rangle$ → f is constant.
- Measure anything else → f is balanced.

A concrete 2-bit example

Let's take $n = 2$ and a balanced function:
$f(00) = 0,\ f(01) = 1,\ f(10) = 1,\ f(11) = 0$ (two 0's, two 1's).

The state after the oracle is:

$$\frac{1}{2}\big(+ |00\rangle - |01\rangle - |10\rangle + |11\rangle\big)|-\rangle$$

Apply $H^{\otimes 2}$:

- $H^{\otimes 2}|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$
- $H^{\otimes 2}|01\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
- $H^{\otimes 2}|10\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$
- $H^{\otimes 2}|11\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$

Summing with the coefficients $(+1, -1, -1, +1)$:

- Coefficient of $|00\rangle$: $\frac{1}{4}(1 - 1 - 1 + 1) = 0$
- Coefficient of $|01\rangle$: $\frac{1}{4}(1 + 1 - 1 - 1) = 0$

- Coefficient of $|10\rangle$: $\frac{1}{4}(1-1+1-1)=0$
- Coefficient of $|11\rangle$: $\frac{1}{4}(1+1+1+1)=1$

So the state becomes $|11\rangle|-\rangle$. Measurement yields $|11\rangle$ (non-zero) → balanced. ✅

Summary – Why is this remarkable?

Algorithm	Classical queries (worst case)	Quantum queries
Deutsch	2	1
Deutsch–Jozsa	$2^{n-1}+1$	1

The quantum solution uses **superposition** and **phase kickback** to evaluate all inputs simultaneously, then **interference** (via Hadamard) to extract the global property "constant or balanced" in a single measurement.

Bernstein–Vazirani Algorithm

In Deutsch-Jozsa algorithm,

- Measure $|0\ldots0\rangle$ → f is constant.
- Measure anything else → f is balanced.

Can we have more information about the function?

Problem statement

We are given an unknown **hidden string** $s \in \{0,1\}^n$.
A quantum oracle computes the function

$$f(x) = s \cdot x = \bigoplus_{i=1}^{n} s_i x_i \quad \text{(dot product modulo 2)}.$$

Goal: Determine s using as few oracle queries as possible.

Classically, one query gives only one bit $f(x)$; to find all n bits of s you need at least n queries (e.g., query $x = 1, 2, 4, \ldots$).

On quantum circuit, a single query suffices!

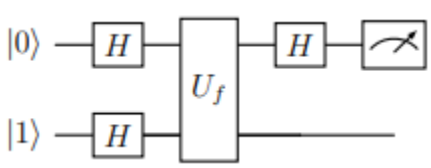
Circuit and phase oracle

Because $f(x) = s \cdot x$, the phase kickback trick gives

$$U_f|x\rangle|-\rangle = (-1)^{s \cdot x}|x\rangle|-\rangle.$$

Thus the oracle effectively applies the phase $(-1)^{s \cdot x}$ to the input register.

The circuit is **identical** to the Deutsch–Jozsa circuit:



A concrete 2-bit example ($s = 10$)

Let $n = 2$ and $s = 10$ (i.e., $s_1 = 1, s_2 = 0$).
Then $f(x) = x_1$ (the first bit).
Truth table:
$f(00) = 0,\ f(01) = 0,\ f(10) = 1,\ f(11) = 1.$

Step-by-step calculation:

1. **Initial state:** $|00\rangle|1\rangle$

2. **Apply $H^{\otimes 2}$ on top, H on bottom:**

$$\frac{1}{2}\big(|00\rangle + |01\rangle + |10\rangle + |11\rangle\big) \otimes |-\rangle$$

3. **Oracle phase kickback:**
   Each $|x\rangle$ gets factor $(-1)^{f(x)}$:

$$\frac{1}{2}\big(+ |00\rangle + |01\rangle - |10\rangle - |11\rangle\big)|-\rangle$$

4. **Apply $H^{\otimes 2}$ to the top register.**
   Instead of doing the full sum, we use the known identity:

$$H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_x (-1)^{s\cdot x}|x\rangle\right) = |s\rangle.$$

   Here the normalisation of our state is $\frac{1}{2}$, which is exactly $\frac{1}{\sqrt{2^n}}$ for $n = 2$.
   So after $H^{\otimes 2}$ the state becomes $|10\rangle$.

5. **Measurement of the top register** yields 10 with certainty.

Thus one run reveals the whole hidden string s.

Why the identity holds: Working with Hadamard gates

For any s:

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle.$$

Therefore,

$$H^{\otimes n} \left(\frac{1}{\sqrt{2^n}} \sum_x (-1)^{s \cdot x} |x\rangle \right) = \frac{1}{2^n} \sum_{x,z} (-1)^{s \cdot x + x \cdot z} |z\rangle = \frac{1}{2^n} \sum_z \left(\sum_x (-1)^{x \cdot (s \oplus z)} \right) |z\rangle.$$

The inner sum is 2^n if $s \oplus z = 0$ (i.e., $z = s$), and 0 otherwise. Hence the result is $|s\rangle$.

Simon's Problem and Algorithm

We are given a function $f : \{0,1\}^n \to \{0,1\}^m$ (usually $m \geq n-1$) with a **promise**:
There exists a **non-zero** string $s \in \{0,1\}^n$ such that

$$f(x) = f(y) \iff y = x \text{ or } y = x \oplus s.$$

In other words, f is **two-to-one** with period s (the case $s = 0$ would be one-to-one, but we exclude it).
Goal: Find the hidden period s using as few queries as possible.

Classically, you need to find a collision $f(x) = f(x \oplus s)$. The best classical algorithm takes $\Omega(2^{n/2})$ queries (birthday bound).

Quantum; Simon's algorithm finds s with only $O(n)$ queries – an exponential speedup.

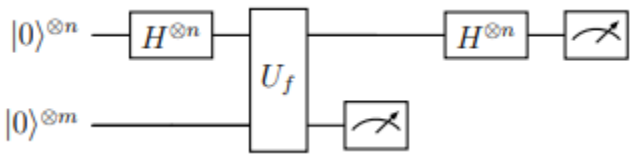
Oracle and circuit

The oracle now works on two registers:

$$U_f|x\rangle|0\rangle = |x\rangle|f(x)\rangle.$$

Note: we do **not** use the $|-\rangle$ trick here; instead we use the **second register** to store $f(x)$ and then measure it.

The circuit (repeated $O(n)$ times):



After the first Hadamard we have

$$\frac{1}{\sqrt{2^n}} \sum_x |x\rangle|0\rangle.$$

Next apply oracle

The oracle produces

$$\frac{1}{\sqrt{2^n}} \sum_x |x\rangle|f(x)\rangle.$$

Now we **measure the second register**.

Suppose we obtain a value $f(x_0)$. Because f is two-to-one, the first register collapses to the superposition

$$\frac{1}{\sqrt{2}}\big(|x_0\rangle + |x_0 \oplus s\rangle\big).$$

Remember the useful identity (again)

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle$$

Applying the final Hadamard

Apply $H^{\otimes n}$ to the first register. Using the identity $H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle$:

$$H^{\otimes n}\left(\frac{1}{\sqrt{2}}(|x_0\rangle + |x_0 \oplus s\rangle)\right) = \frac{1}{\sqrt{2^{n+1}}} \sum_z \left((-1)^{x_0 \cdot z} + (-1)^{(x_0 \oplus s) \cdot z}\right)|z\rangle.$$

Now $(x_0 \oplus s) \cdot z = x_0 \cdot z \oplus s \cdot z$. Hence:

$$(-1)^{x_0 \cdot z} + (-1)^{x_0 \cdot z \oplus s \cdot z} = (-1)^{x_0 \cdot z}\big(1 + (-1)^{s \cdot z}\big).$$

- If $s \cdot z = 0 \pmod 2$, the bracket is $1 + 1 = 2$.
- If $s \cdot z = 1$, the bracket is $1 - 1 = 0$.

Therefore the final state has **non-zero amplitude only for those z such that $s \cdot z = 0$.** Moreover, all such z appear with equal probability.

When we measure the first register, we obtain a random z satisfying $z \cdot s = 0$.

After many runs, we collect several z values that satisfy $z \cdot s = 0$.

Solve equation to find s.

A concrete 2-bit example ($s = 11$)

Take $n = 2$, $s = 11$. Define f as:

$f(00) = a,\ f(11) = a$
$f(01) = b,\ f(10) = b$
with $a \neq b$.

Step-by-step:

1. **Initial:** $|00\rangle|0\rangle$ (the second register has enough qubits to hold a, b; we don't need to specify m).

---

2. **Hadamard on first register:**

$$\frac{1}{2}\big(|00\rangle + |01\rangle + |10\rangle + |11\rangle\big)|0\rangle$$

---

3. **Oracle:**

$$\frac{1}{2}\big(|00\rangle|a\rangle + |01\rangle|b\rangle + |10\rangle|b\rangle + |11\rangle|a\rangle\big)$$

---

4. **Measure second register.**
   - If we get a: first register collapses to $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
   - If we get b: collapses to $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$.

---

5. **Apply $H^{\otimes 2}$ to the first register.**
   - For the state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

$$H^{\otimes 2}|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$H^{\otimes 2}|11\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$$

Sum = $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: this means we only get $z = 00$ or 11

Verification: the condition $s \cdot z = 0$: for $s = 11$, $z_1 + z_2 = 0 \mod 2 \Rightarrow z_1 = z_2$. So allowed z are indeed 00 and 11.

- For the other branch $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$:
  $H^{\otimes 2}|01\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
  $H^{\otimes 2}|10\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$
  Sum = $\frac{1}{2\sqrt{2}}[2|00\rangle + 0|01\rangle + 0|10\rangle - 2|11\rangle] = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$.
  Measuring yields either $|00\rangle$ or $|11\rangle$ with equal probability.

Interpretation: After many runs, we collect several z values that satisfy $z \cdot s = 0$.

- For $n = 2$, the only non-trivial constraint comes from $z = 11$:
  - $11 \cdot s = s_1 + s_2 = 0 \Rightarrow s_1 = s_2$.
- Since $s \neq 00$; (in Simon's problem, the promise is that there exists a **non-zero string** s), we deduce $s = 11$.

The full Simon's algorithm analysis

Repeat the above procedure $O(n)$ times. Each run gives a random z (non-zero with good probability) satisfying $z \cdot s = 0$. Collect enough linearly independent z (over $\mathbb{F}_2$) to form a system of equations that determines s uniquely. Solve (e.g., by Gaussian elimination) to recover s.

Query complexity: Each run uses one oracle call, and $O(n)$ runs suffice → exponential speedup over classical $\Omega(2^{n/2})$.

Summary of all four algorithms

Algorithm	Problem	Classical queries	Quantum queries
Deutsch	1-bit constant vs balanced	2	1
Deutsch–Jozsa	n-bit constant vs balanced	$2^{n-1}+1$	1
Bernstein–Vazirani	Find hidden dot-product string	n	1
Simon	Find hidden period of a 2-to-1 function	$\Theta(2^{n/2})$	$O(n)$

Each algorithm uses superposition and interference to extract global properties of a function with exponentially fewer queries than the best classical method.

This completes the set of foundational quantum algorithms. You are now ready to understand Shor's factoring algorithm, which builds on the same principles (phase estimation, period finding).

---

1. **Deutsch algorithm** – Work out the case where $f(0)=1, f(1)=0$ (balanced). Show that the final measurement yields $|1\rangle$.
2. **2-bit constant** – Repeat the 2-bit calculation for a constant function, e.g. $f(x)=0$ for all x. Verify that the final state is $|00\rangle|-\rangle$.
3. **Why the $|-\rangle$ state?** – Try running the Deutsch–Jozsa circuit with the bottom qubit starting in $|0\rangle$ instead of $|1\rangle$. Why does it fail?
4. **Balanced function for $n=2$** – Choose $f(00)=0, f(01)=1, f(10)=0, f(11)=1$ (balanced). Simulate the Deutsch-Jozsa circuit step-by-step (by hand or with PennyLane) and show that you never measure $|00\rangle$.
5. **Bernstein-Vazirani** – For $n=3$, hidden string $s=101$, compute $f(x)=s\cdot x$ for all x and then run the quantum circuit (conceptually) to see why the output is $|101\rangle$.
6. **Bernstein–Vazirani** – Work out the $n=3$ case for $s=101$. Show that after the final Hadamard you measure 101.

7. **Simon's algorithm** – Suppose $n = 3$ and $s = 110$. List all possible z values that can be measured. Why is it impossible to get $z = 001$?
8. **Why measurement?** – In Simon's algorithm, why do we measure the second register before the final Hadamard? What would happen if we didn't measure it?
9. **Classical vs quantum** – For $n = 20$, how many classical queries are needed on average to find a collision? How many quantum queries does Simon's algorithm use? Compute the ratio.
10. **AI exercise** – Use an AI to generate a constant oracle for $n = 4$ and a balanced oracle. Then write a PennyLane script that runs Deutsch-Jozsa on both and prints the measurement outcomes.
11. **Challenge** – Implement Simon's algorithm for $n = 2$ with period $s = 01$ in PennyLane.

Next Lecture Preview

Shor's integer factoring algorithm